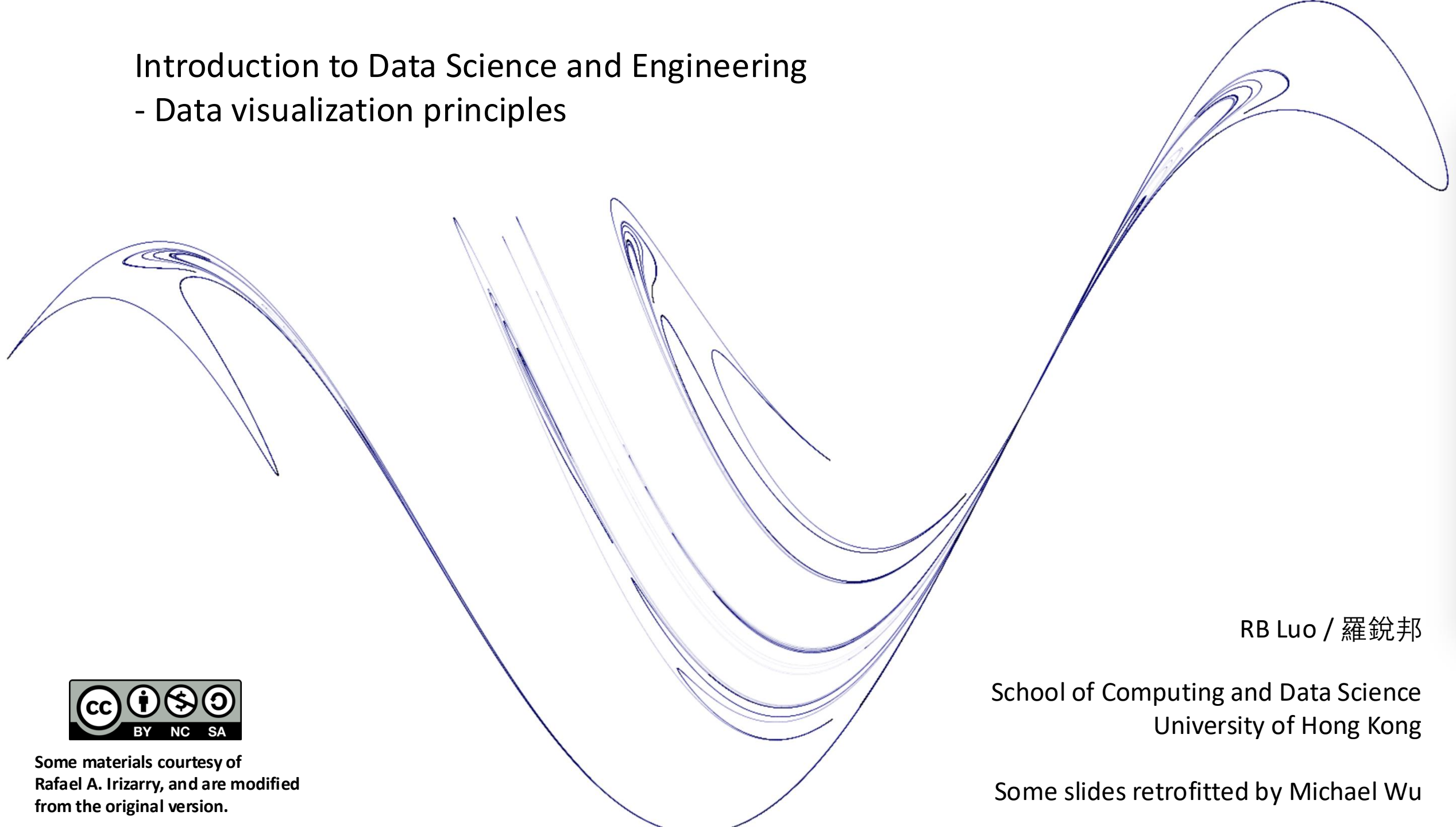


Introduction to Data Science and Engineering

- Data visualization principles



RB Luo / 羅銳邦

School of Computing and Data Science
University of Hong Kong

Some slides retrofitted by Michael Wu



Some materials courtesy of
Rafael A. Irizarry, and are modified
from the original version.



Next class is a lab

- Please bring your own laptop, please make sure RStudio is installed and functional on your laptop
- Attendance is taken for the two labs
- You will be given a lab sheet, you are expected to answer the questions in the sheet in class. You are encouraged to try different code or arguments. You can discuss with other students for ideas but do not copy from others
- If you have any questions, you can ask Google, the TAs, or me
- Sample answers will be given after the class



We have learned to generate plots with a few examples

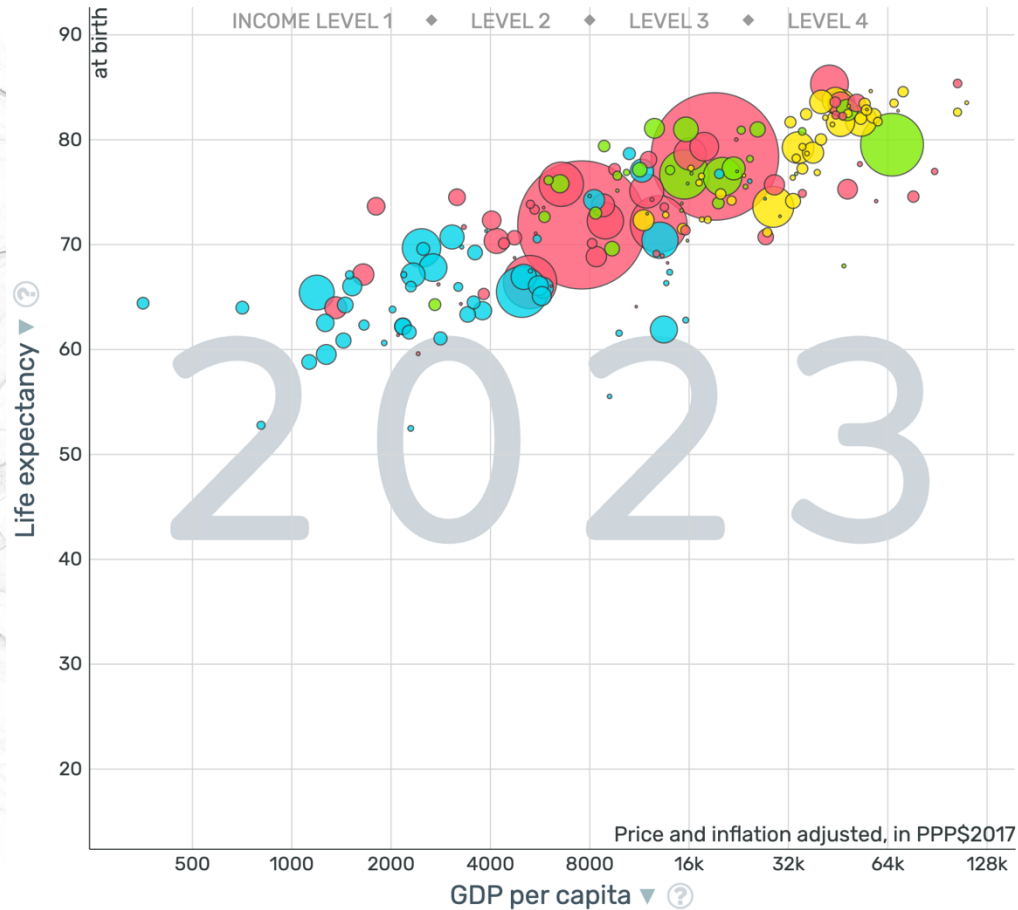
In this lecture we show some examples of plot styles we should avoid, explain how to improve them, and use these as motivation for a list of principles

The principles are mostly based on research related to how humans detect patterns and make visual comparisons. The preferred approaches are those that best fit the way our brains process visual information

When deciding on a visualization approach, it is also important to keep our goal in mind. We may be

- Comparing a viewable number of quantities
- Describing distributions for categories or numeric values
- Comparing the data from two groups
- Describing the relationship between two variables

What can we use for visualization?



X-axis:

position / length

Y-axis:

position / length

Movement

Shape:

O or X

Size:

o or O

Angle/orientation:

↑ or →

Patterns/line styles

Color (hue):

red or blue

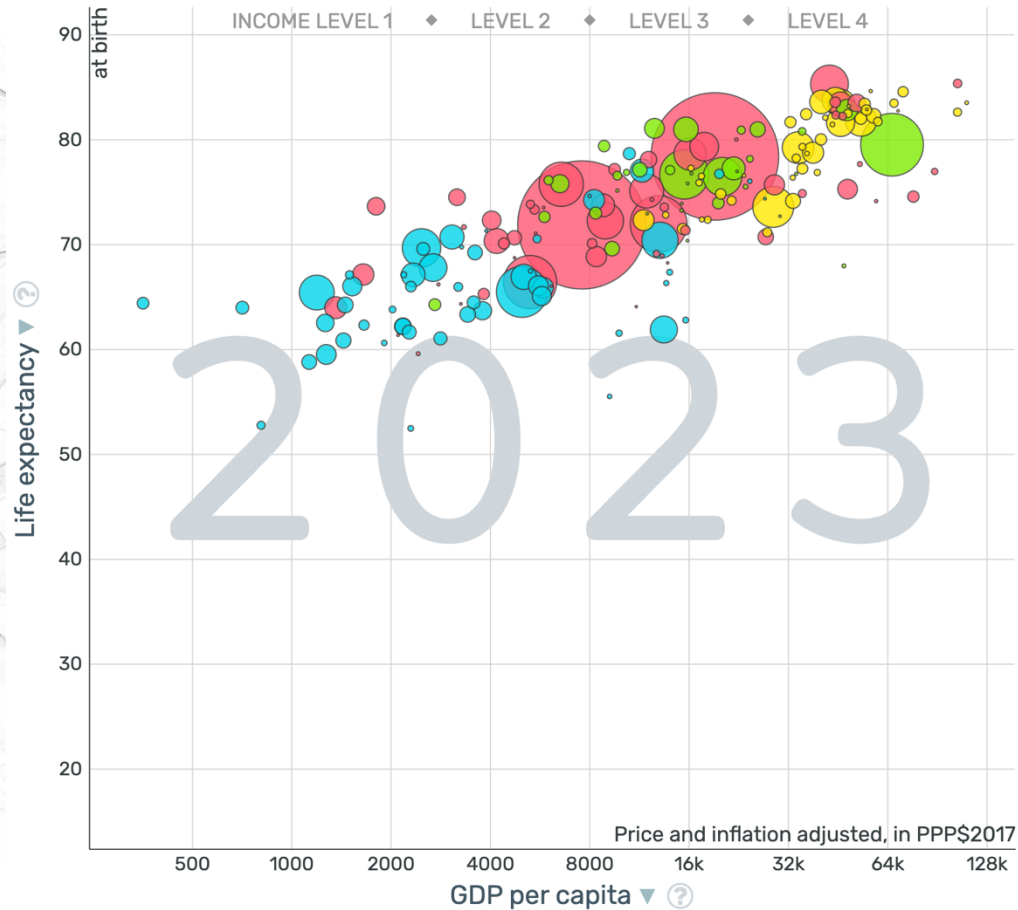
Color (shade):

light or dark

Color (opacity):

opaque or trans

What can we use for visualization?



For numerical variables:

X/Y (position/length), motion >>
size, angle/orientation >
color (shade), color(opacity)

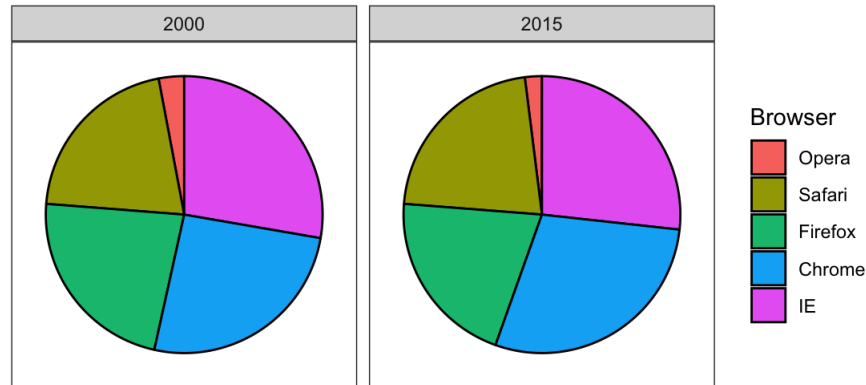
For categorical variables:

X/Y (position/length) >
Shape, patterns/line styles, color(hue) >
Size, angle/orientation, color (shade)

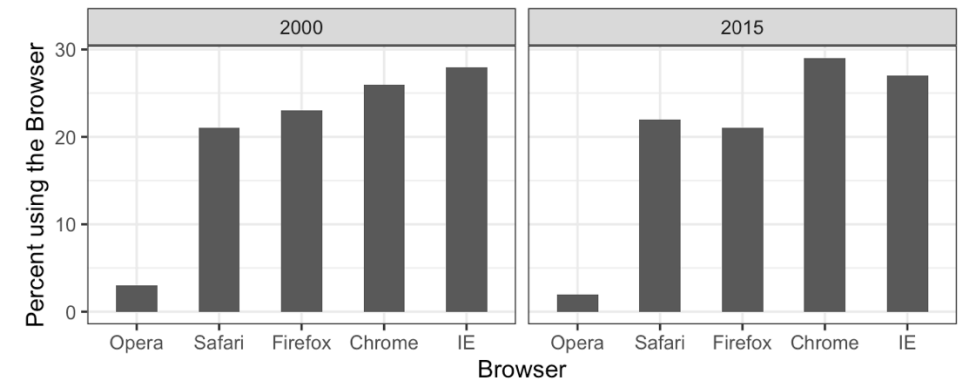
Encoding data using visual cues

To encode data in a plot, we can use position, aligned length, angle, area, brightness, and color hue
Pie chart uses only angle and area, is it a good visualization approach?

Browser	2000	2015
Opera	3	2
Safari	21	22
Firefox	23	21
Chrome	26	29
IE	28	27



VS.



Note

Pie charts are a very bad way of displaying information. The eye is good at judging linear measures and bad at judging relative areas. A bar chart or dot chart is a preferable way of displaying this type of data.

Cleveland (1985), page 264: "Data that can be shown by pie charts always can be shown by a dot chart. This means that judgements of position along a common scale can be made instead of the less accurate angle judgements." This statement is based on the empirical investigations of Cleveland and McGill as well as investigations by perceptual psychologists.

Sales often use pie chart to deliver just only two messages: "we are much better" and "you don't need to care about the numbers"

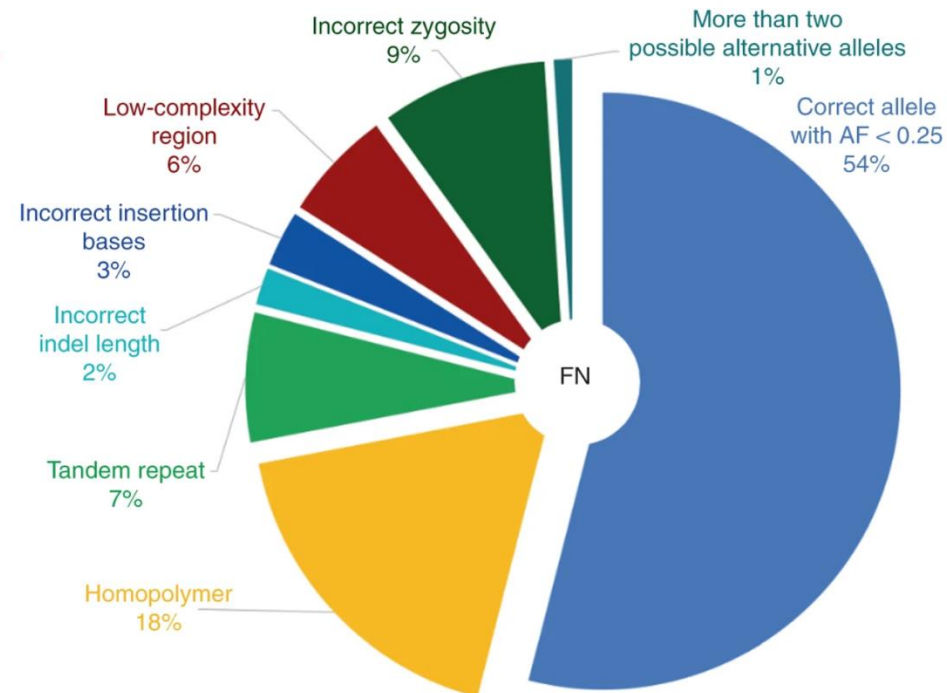
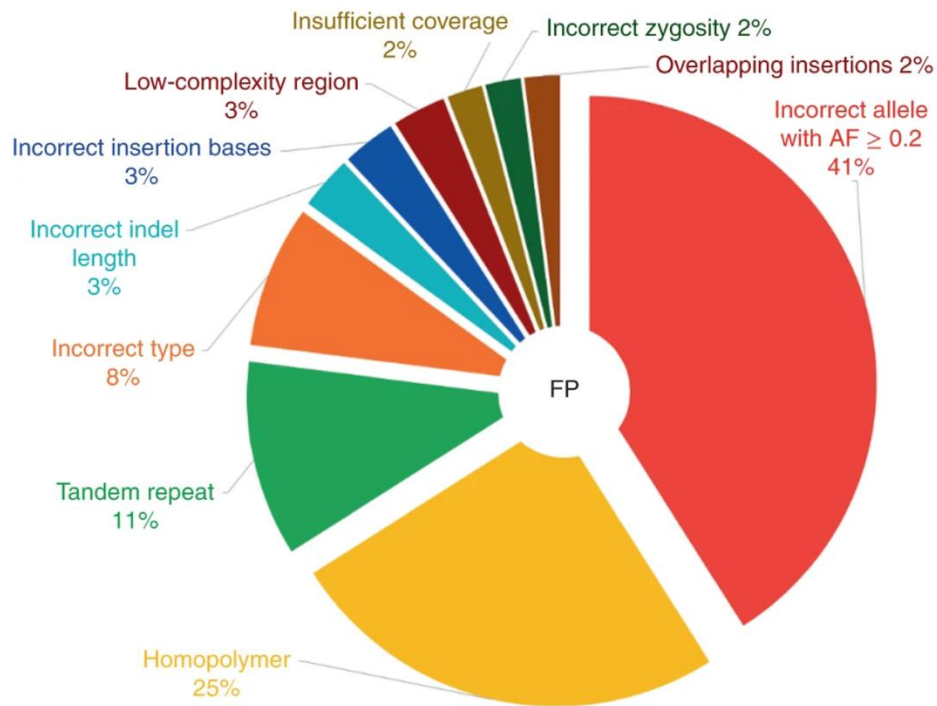
R 4.2.2 · ~/

> ?pie

If for some reasons you need to make a pie chart

Fig. 3: The category distribution of FPs and FNs made by Clair in the 1:168×|2:64× experiment on ONT data and six genome browser screen captures showing examples of different categories.

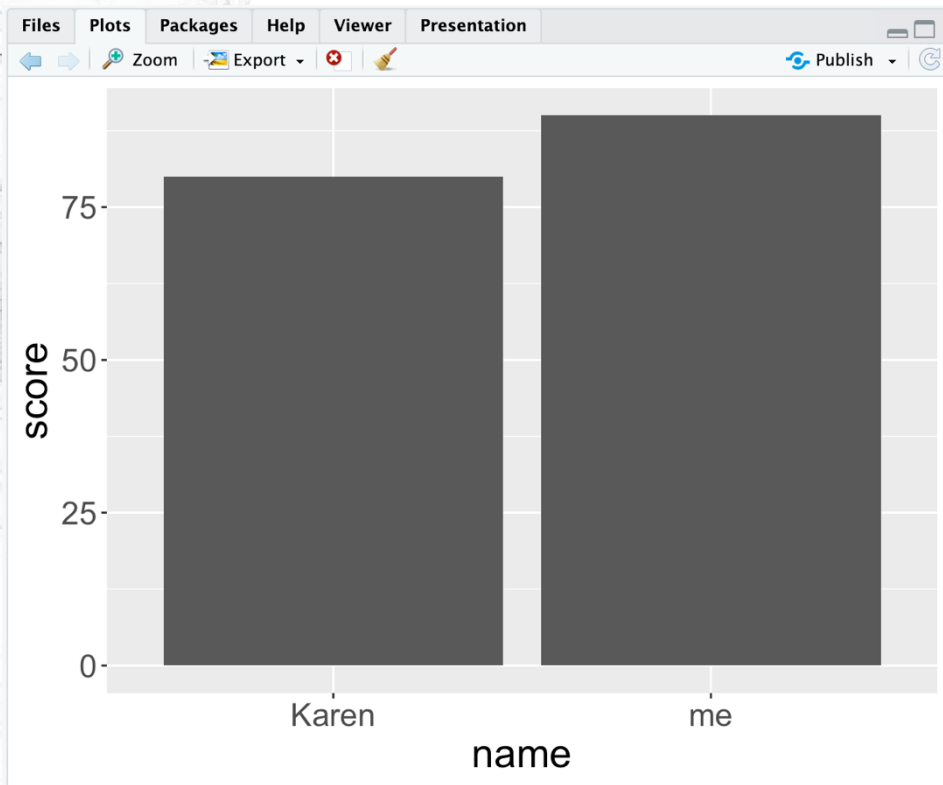
From: [Exploring the limit of using a deep neural network on pileup data for germline variant calling](#)



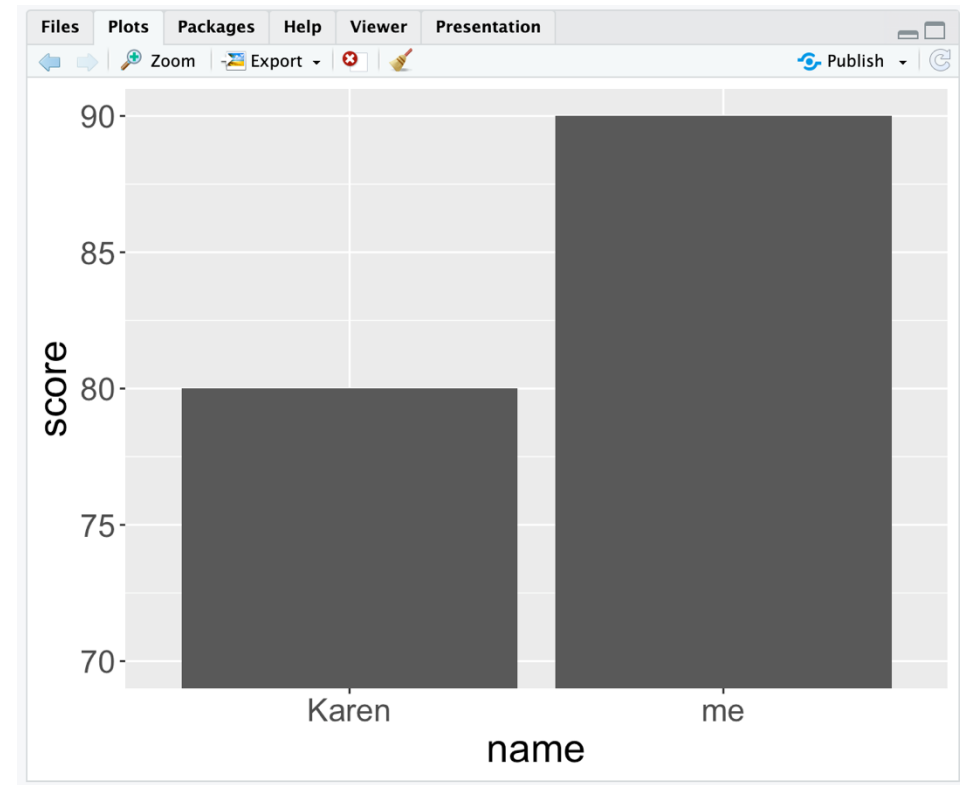
Provide labels and numbers

Know when to include 0

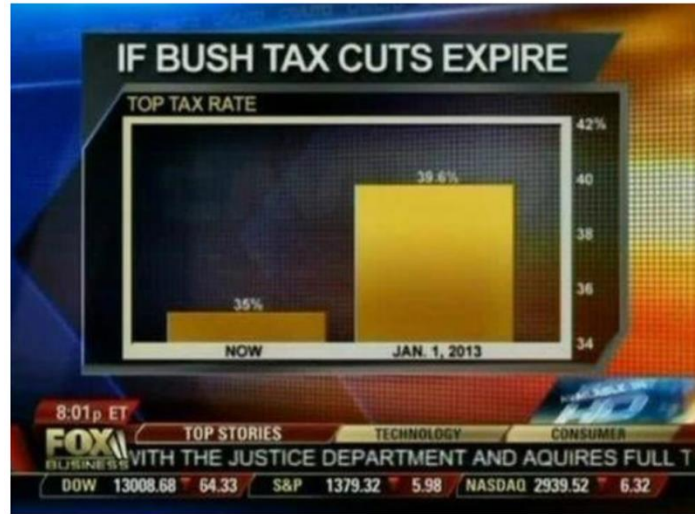
```
R 4.2.2 · ~/   
> df <- data.frame(name=c("me", "Karen"),  
+                  score=c(90,80))  
> ggplot(data=df, aes(x=name, y=score)) +  
+ geom_col() + theme(text = element_text(size=20))
```



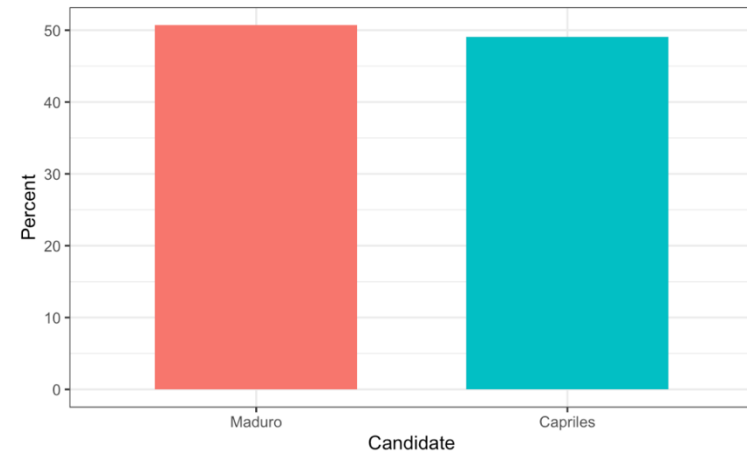
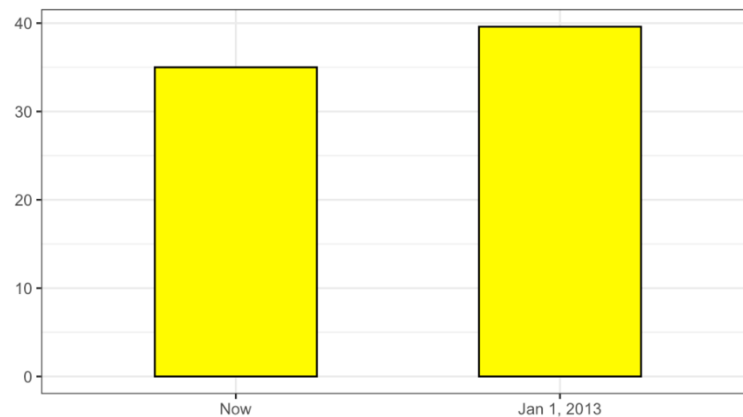
```
R 4.2.2 · ~/   
> ggplot(data=df, aes(x=name, y=score)) +  
+ geom_col() + theme(text = element_text(size=20)) +  
+ coord_cartesian(ylim=c(70, 90))
```



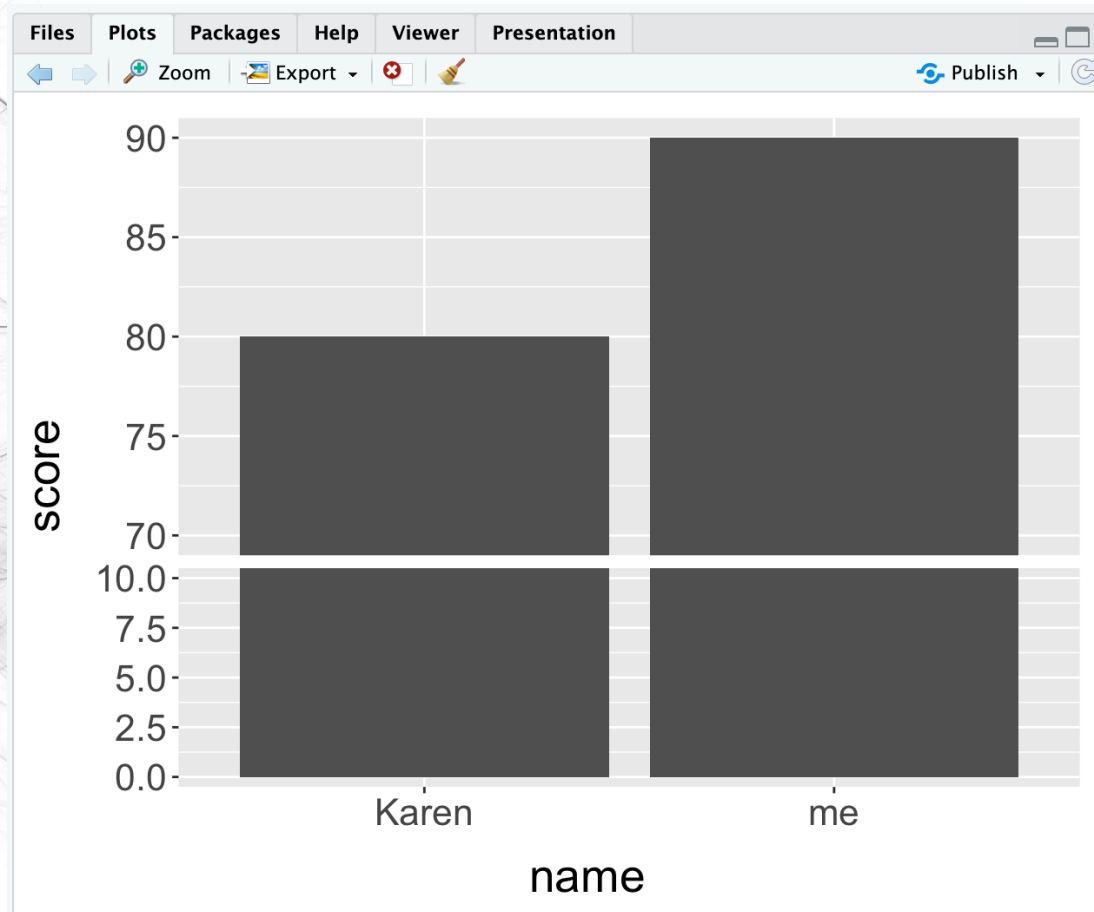
Know where to put 0



Top Tax Rate If Bush Tax Cut Expires



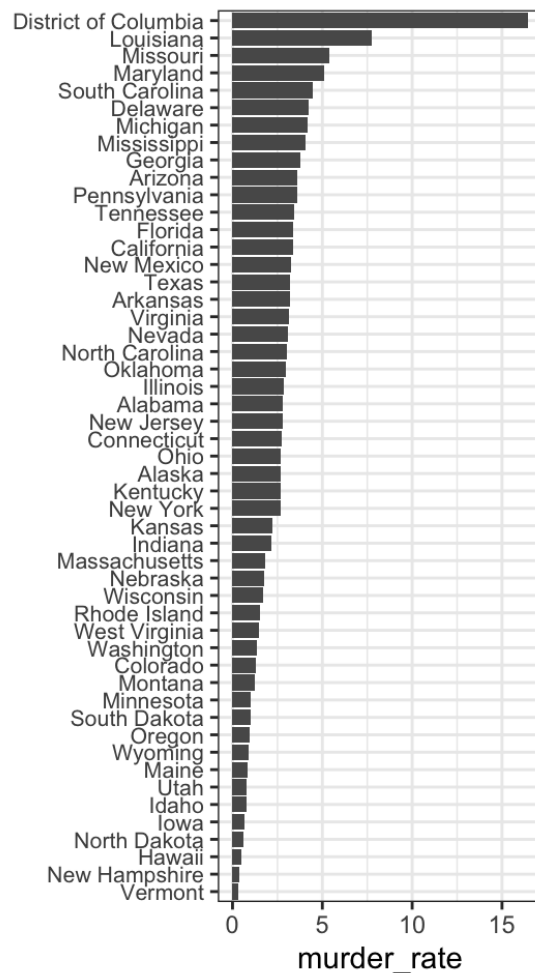
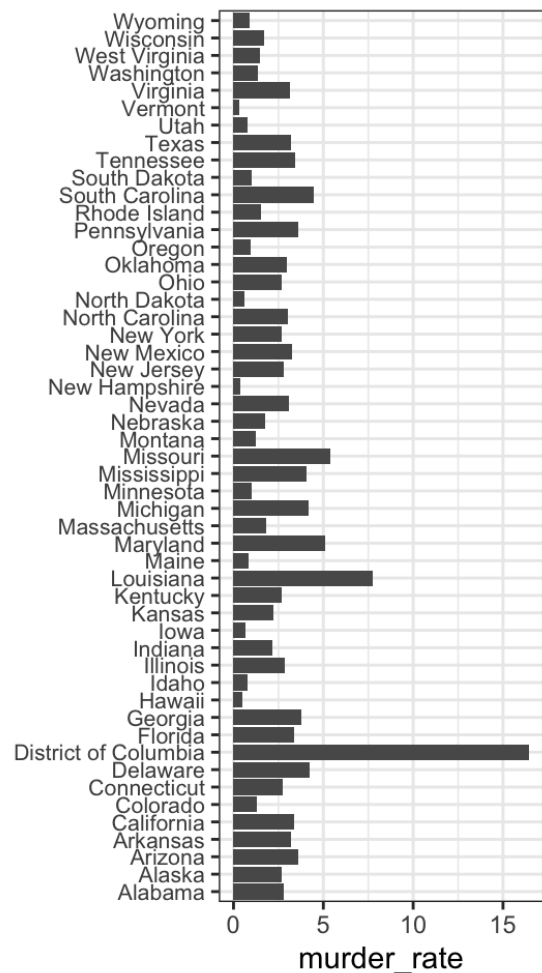
Combine the best of two world on including 0



Use the ggbreak package

```
R 4.2.2 · ~/   
> library(ggbreak)  
> ggplot(data=df, aes(x=name, y=score)) +  
+ geom_col() + theme(text = element_text(size=20)) +  
+ coord_cartesian(ylim=c(0, 90)) + scale_y_break(c(10,70))  
.
```

Order categories by a meaningful value



R 4.2.2 · ~/

```
> data(murders)
> murders |> mutate(murder_rate = total / population * 100000) |>
+ mutate(state = reorder(state, murder_rate)) |>
+ ggplot(aes(state, murder_rate)) +
+ geom_bar(stat="identity") +
+ coord_flip() +
+ theme(axis.text.y = element_text(size = 6)) +
+ xlab("")
```

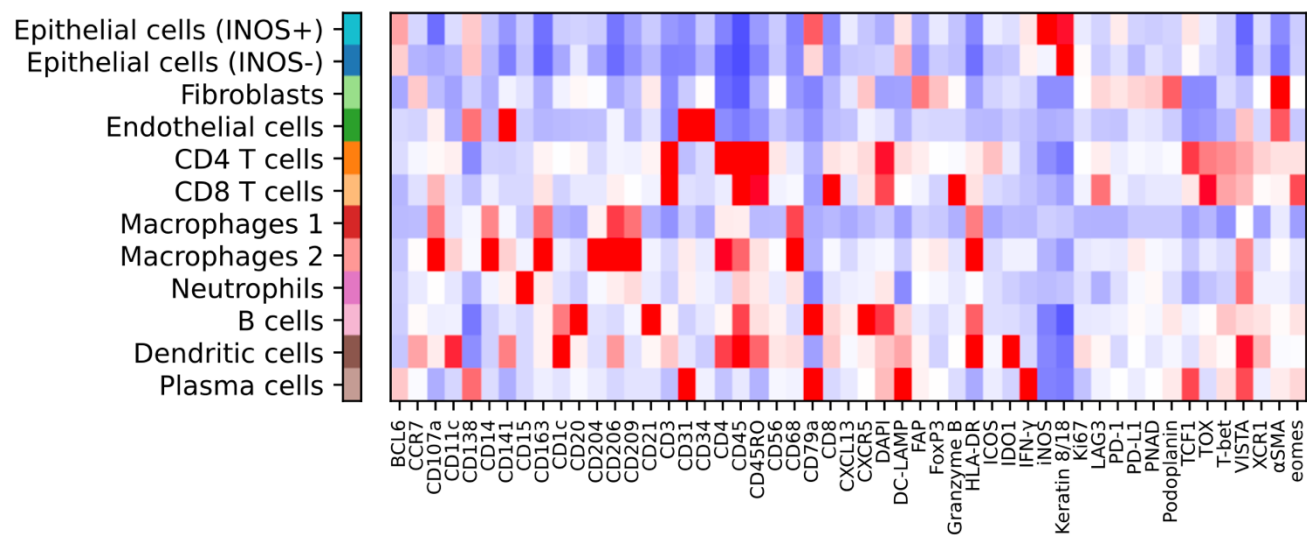
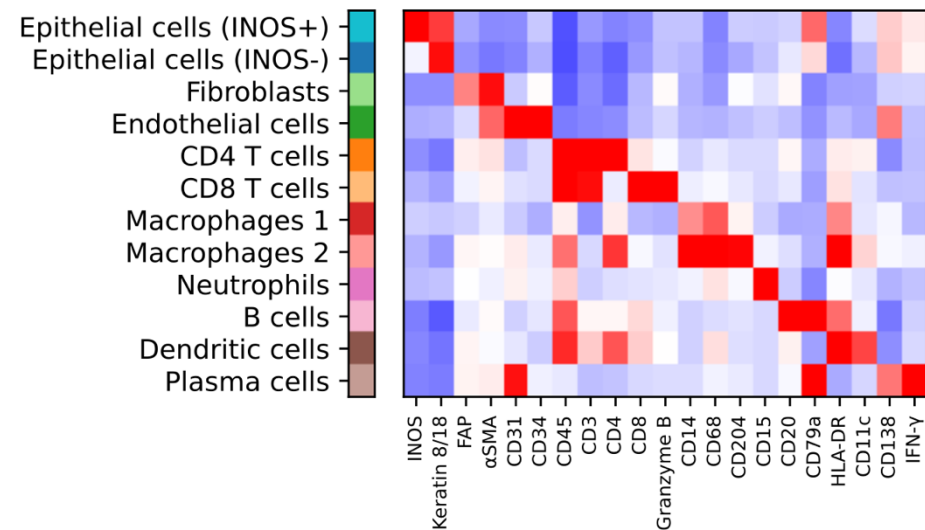
We previously learned how to use the `reorder` function, which helps us achieve this goal. To appreciate how the right order can help convey a message, suppose we want to create a plot to compare the murder rate across states. We are particularly interested in the most dangerous and safest states

Order your categories meaningfully

The two plots here showed protein expression in different types of cells:

- **Red: enriched**
- **Blue: depleted**

Which one makes it easier to spot the signature proteins for each cell type?

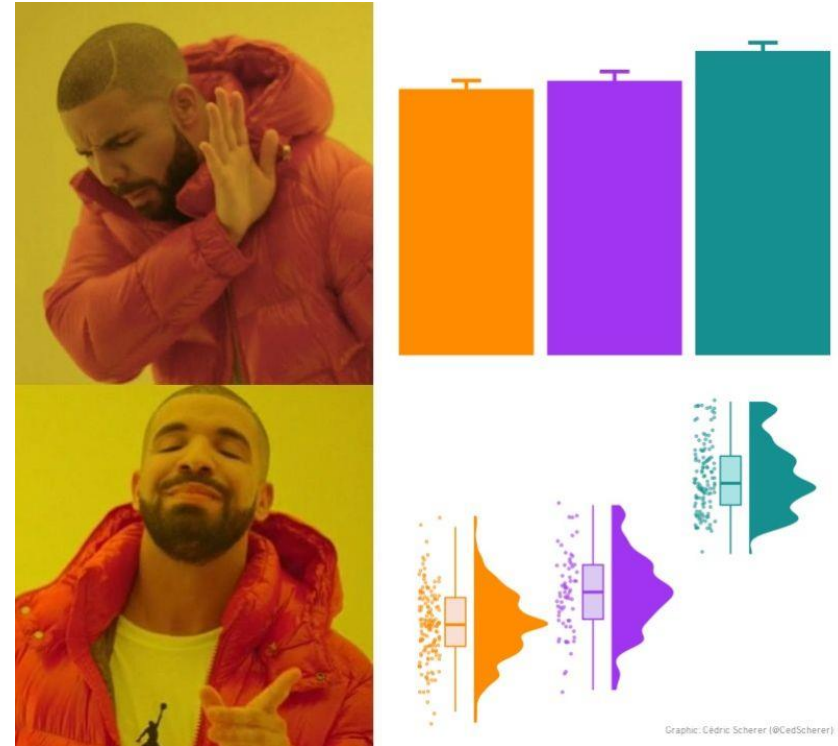


Show your data

Dynamite plot: bar plot with error bar, suggesting that it summarizes multiple numbers.

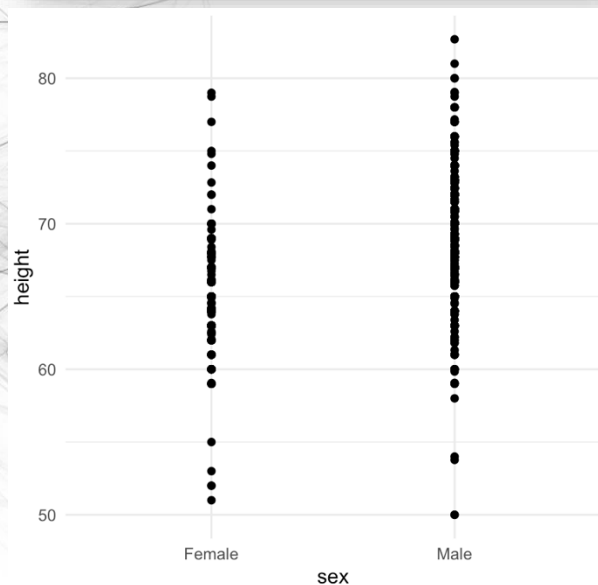
To visualize multiple numbers:

- **Dynamite plot:** mean + SD
- **Box plot:** quartiles, minimum/maximum, outliers
- **Violin/density plot:** full distribution
- **Scatter/swarm plot:** all data points

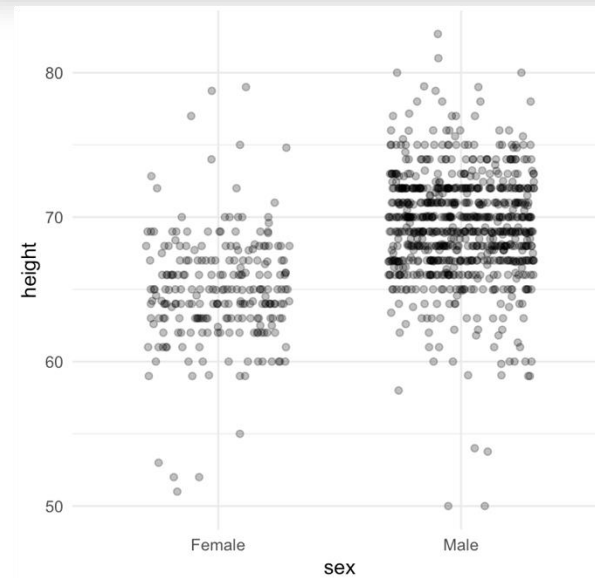


Show your data

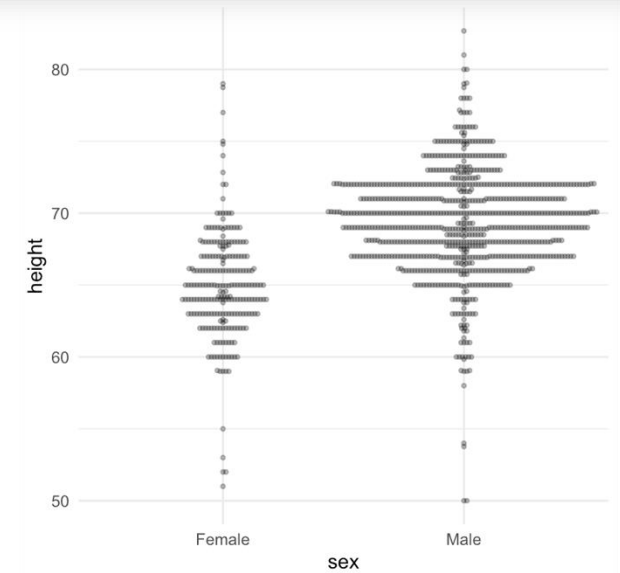
```
> heights |>  
+   ggplot(aes(sex, height)) +  
+   geom_point() +  
+   theme_minimal()
```



```
> heights |>  
+   ggplot(aes(sex, height)) +  
+   geom_jitter(width=0.3, alpha=0.3) +  
+   theme_minimal()
```

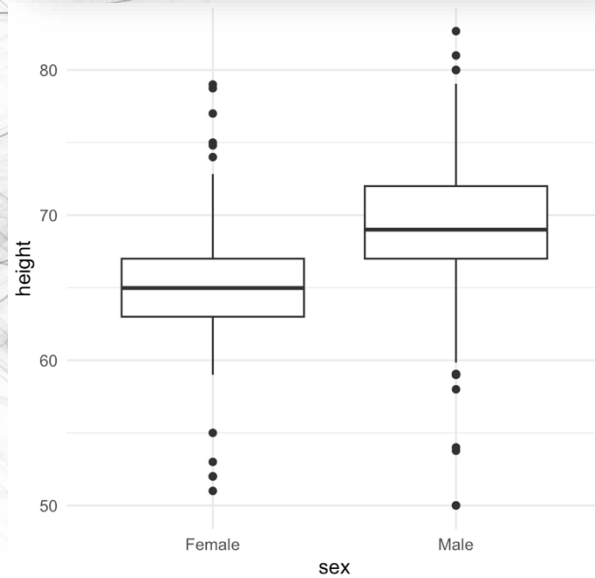


```
> library(ggbeeswarm)  
> heights |>  
+   ggplot(aes(sex, height)) +  
+   geom_beeswarm(alpha=0.3, cex=0.6, size=0.8) +  
+   theme_minimal()
```

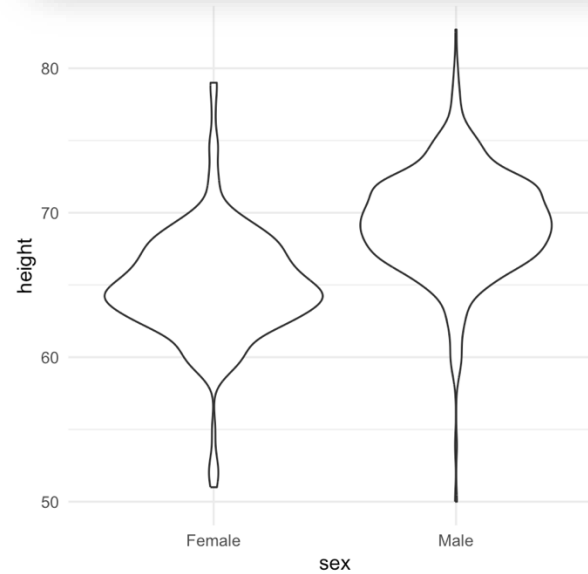


Show your data

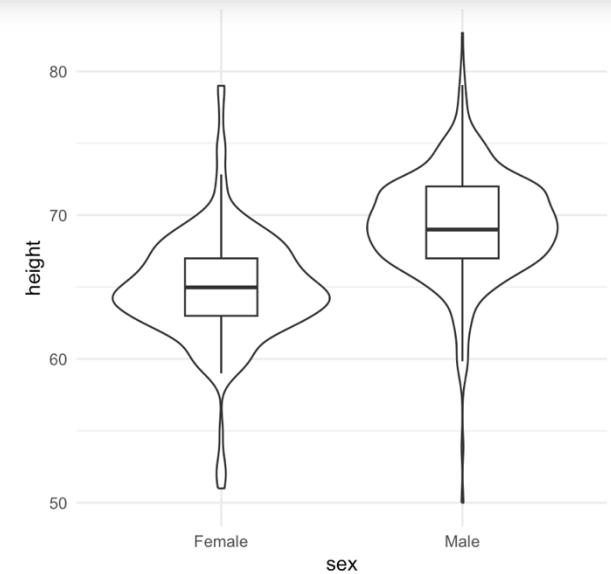
```
> heights |>  
+   ggplot(aes(sex, height)) +  
+   geom_boxplot() +  
+   theme_minimal()
```



```
> heights |>  
+   ggplot(aes(sex, height)) +  
+   geom_violin() +  
+   theme_minimal()
```

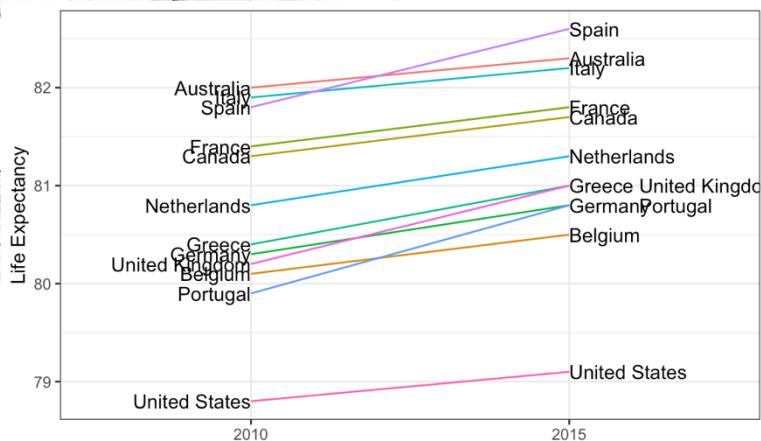


```
> heights |>  
+   ggplot(aes(sex, height)) +  
+   geom_violin() +  
+   geom_boxplot(width=0.3, outliers = FALSE) +  
+   theme_minimal()
```

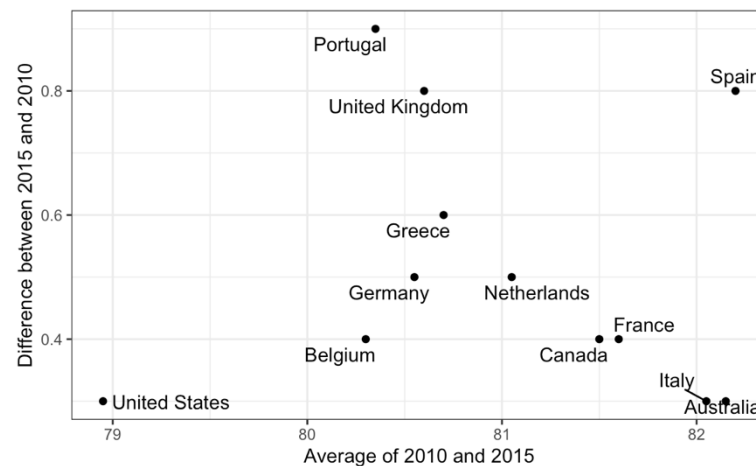


Take note of interesting new plot types in your daily life

Slope chart
Chapter 11.8.1



Bland-Altman plot
Chapter 11.8.2



Home

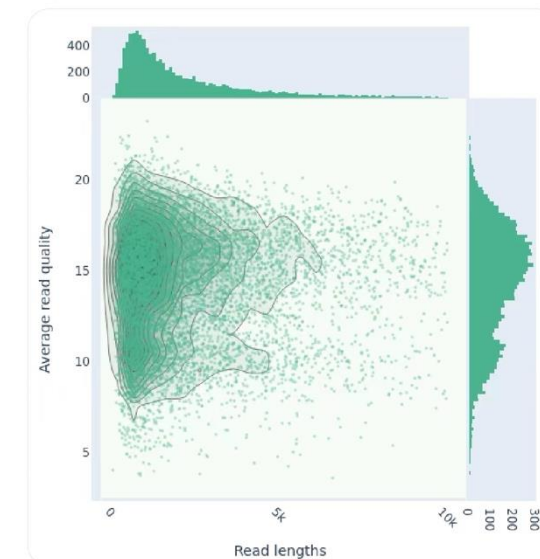
Bioinformatics



Does anyone have any recommendations for genome assembly programs that deal with noisy Nanopore reads with N50s of ~3Kbp at ~100X coverage?

NanoPlot image below. Also have Illumina 150bp paired reads for potential hybrid assembly.

[#AcademicTwitter](#) [#Genomics](#)
[#PromethION](#)



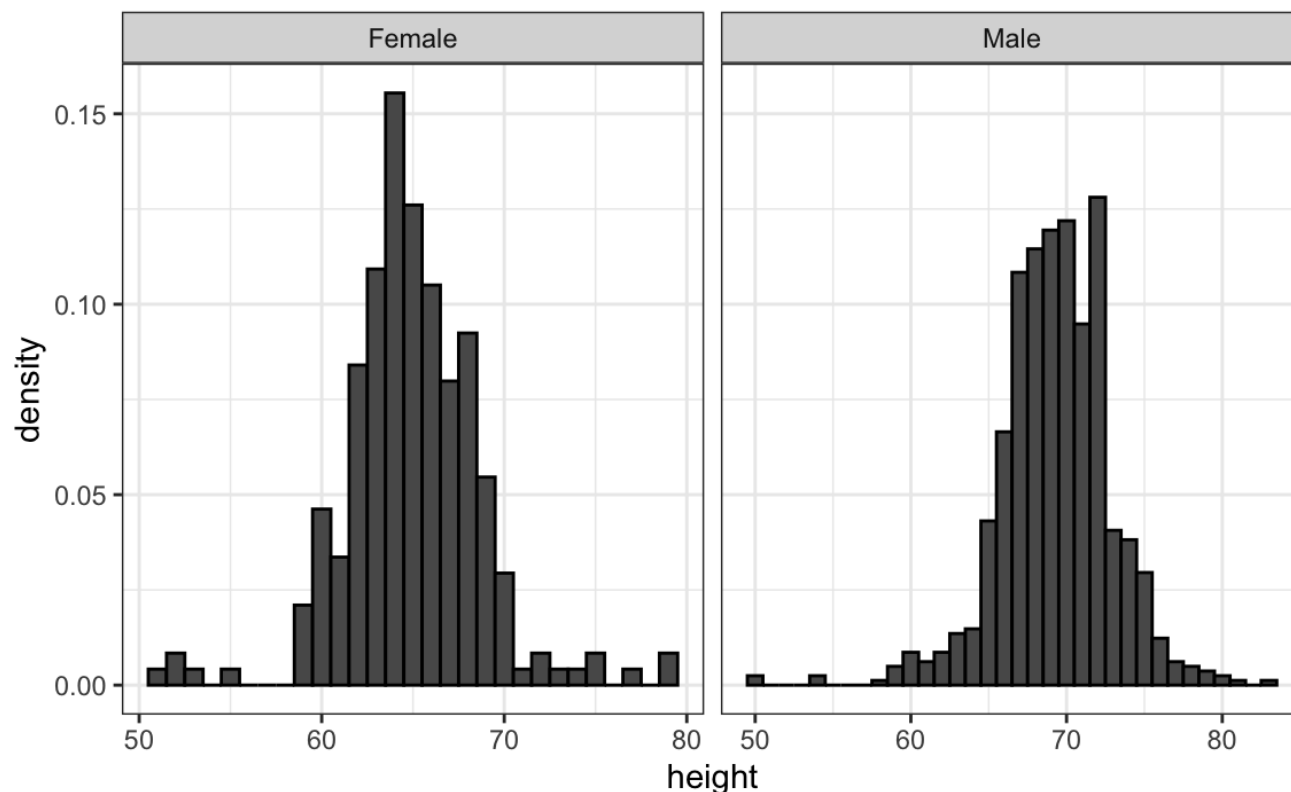
1,756 4 4 6



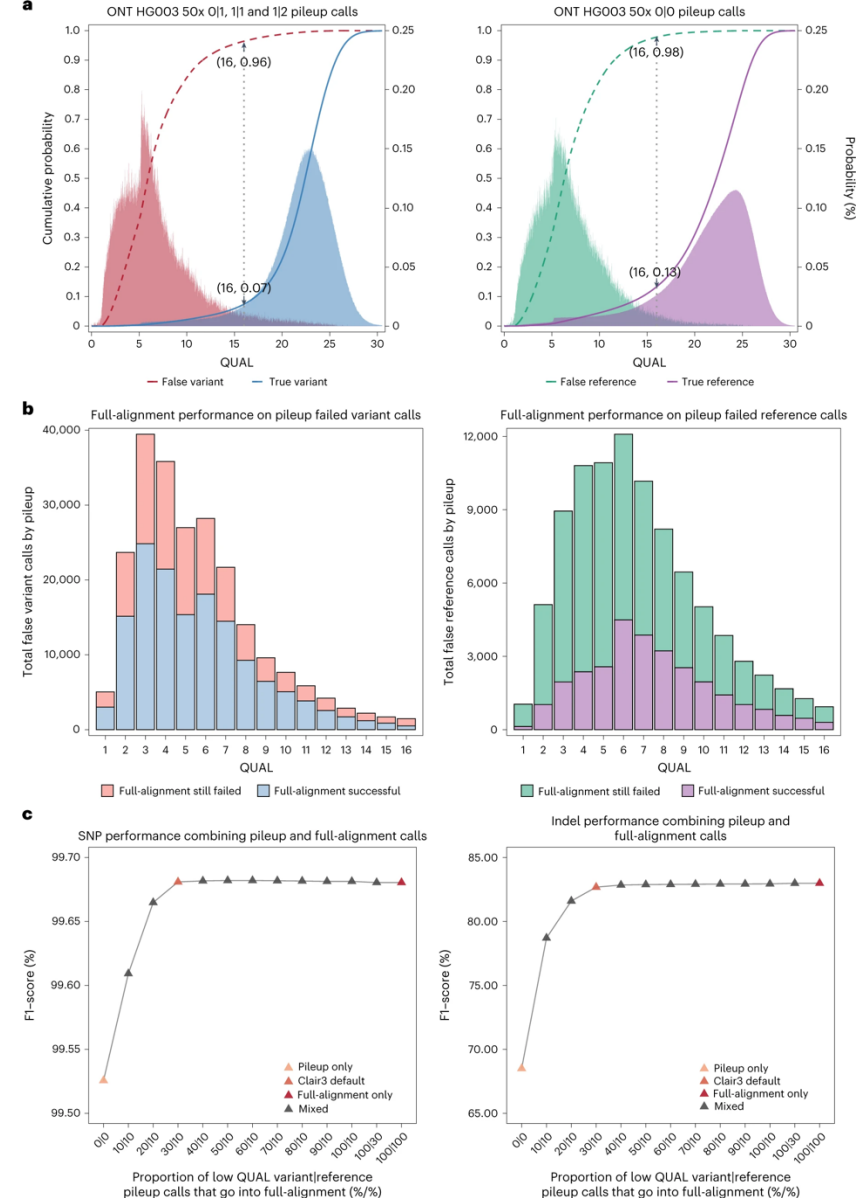
Tuan Nguyen @DrKangabu · 5h
[@Tom_Jenkins](#) what flowcell is this while ago I have great success with Flye for



Use common axes to easy comparison



But only when you intentionally want audience to compare between two plots
Do not over-simplify! Oversimplification is anti-simplicity

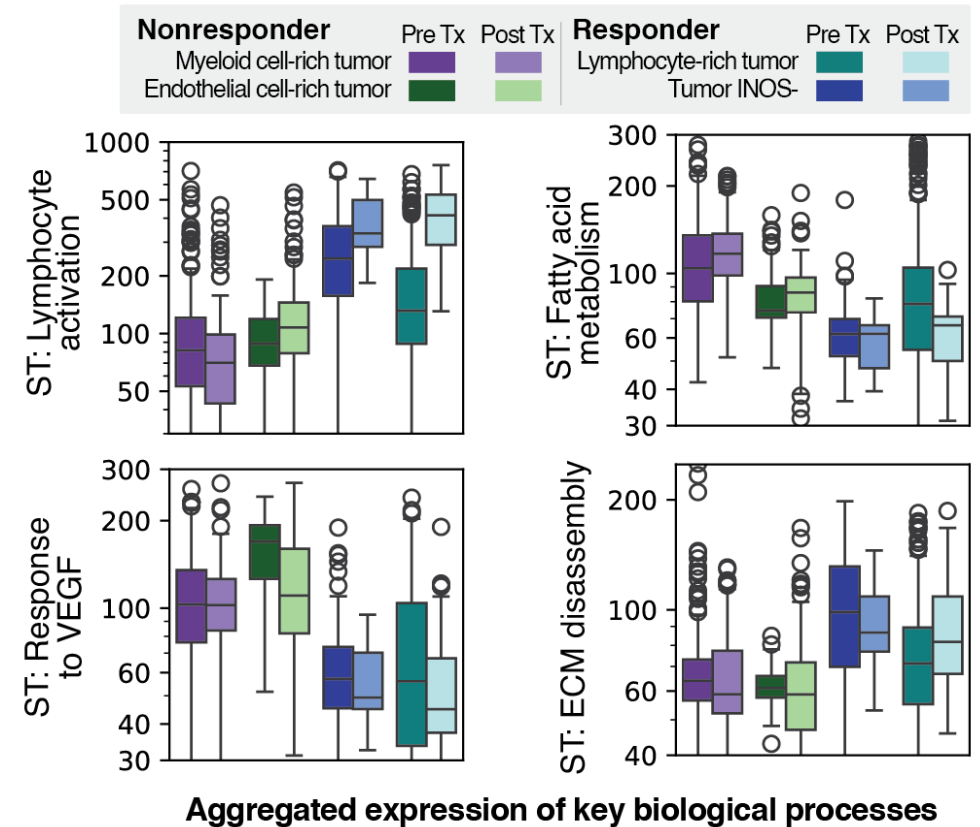


An example that using common
axes is not preferred

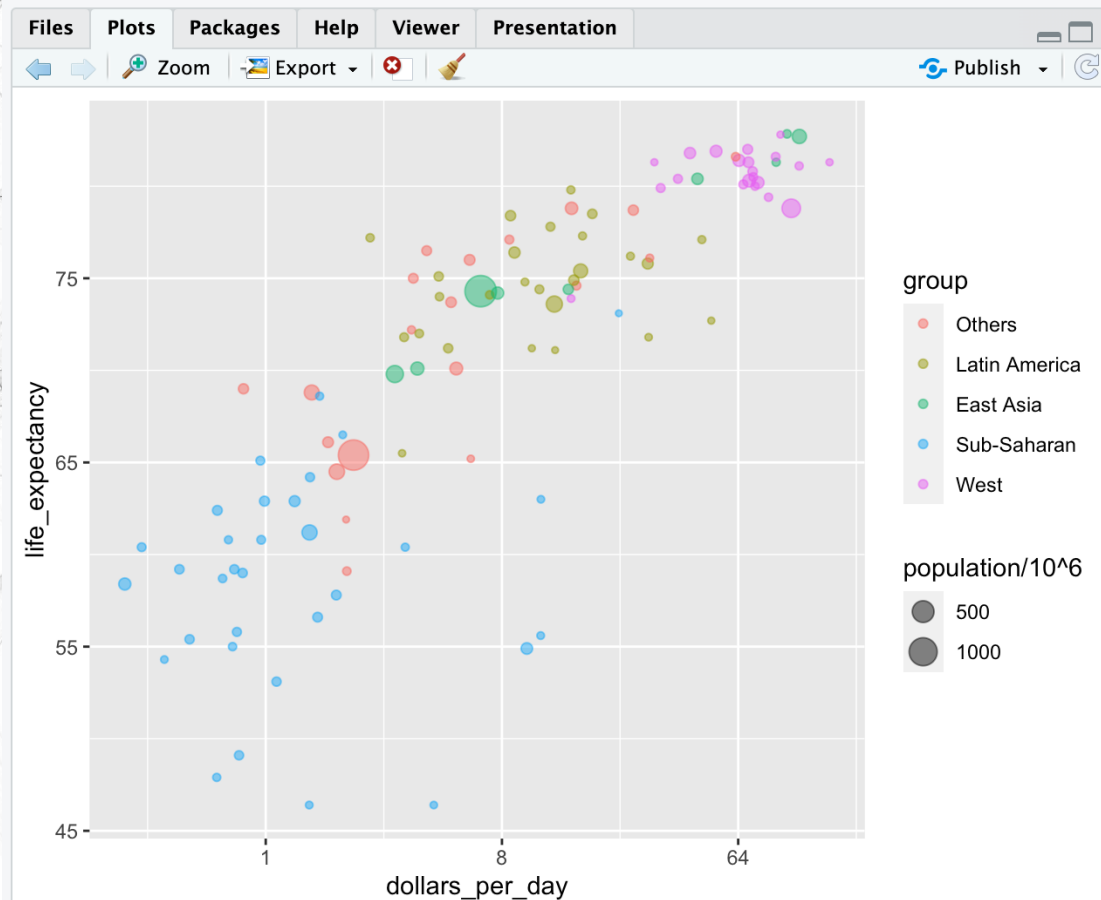
Make good use of colors

- Use no more than six colors in a plot
- Hue versus shade
- Be consistent between plots (use the same palette)
- Use color-blind friendly colors

```
> color_blind_friendly_cols <-  
  c("#999999", "#E69F00", "#56B4E9", "#009E73",  
    "#F0E442", "#0072B2", "#D55E00", "#CC79A7")
```



Adding a third variable

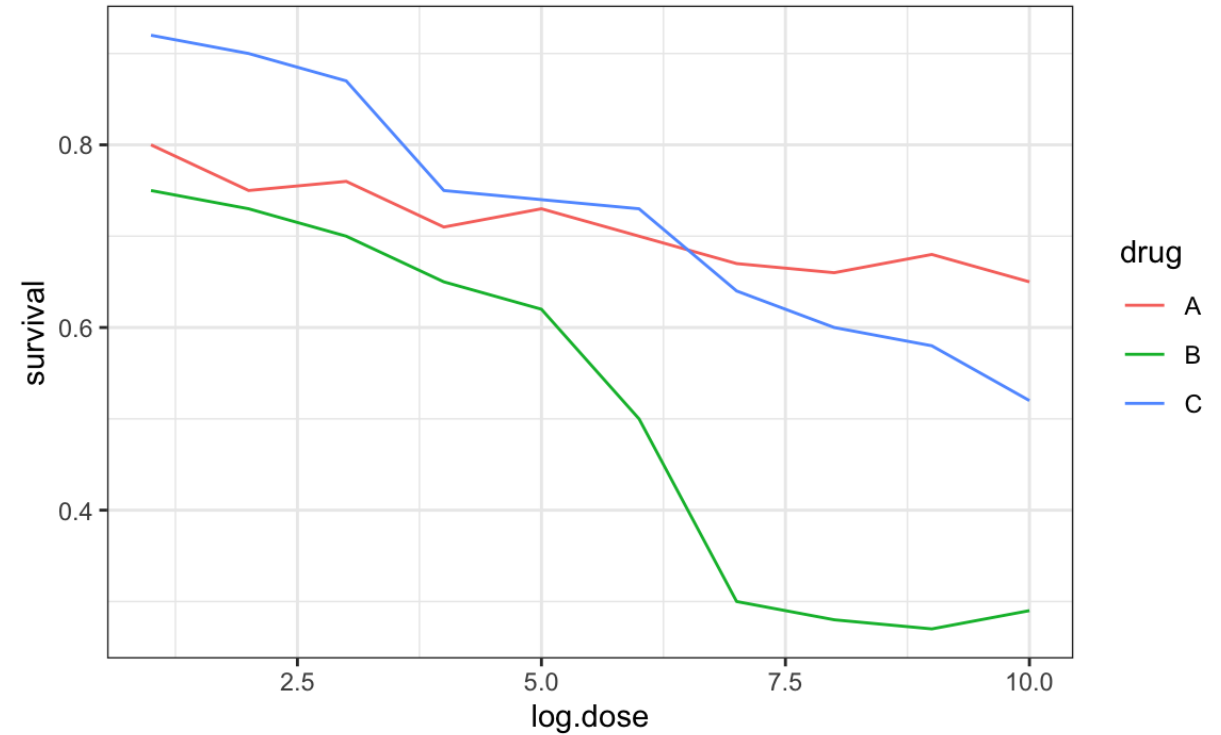
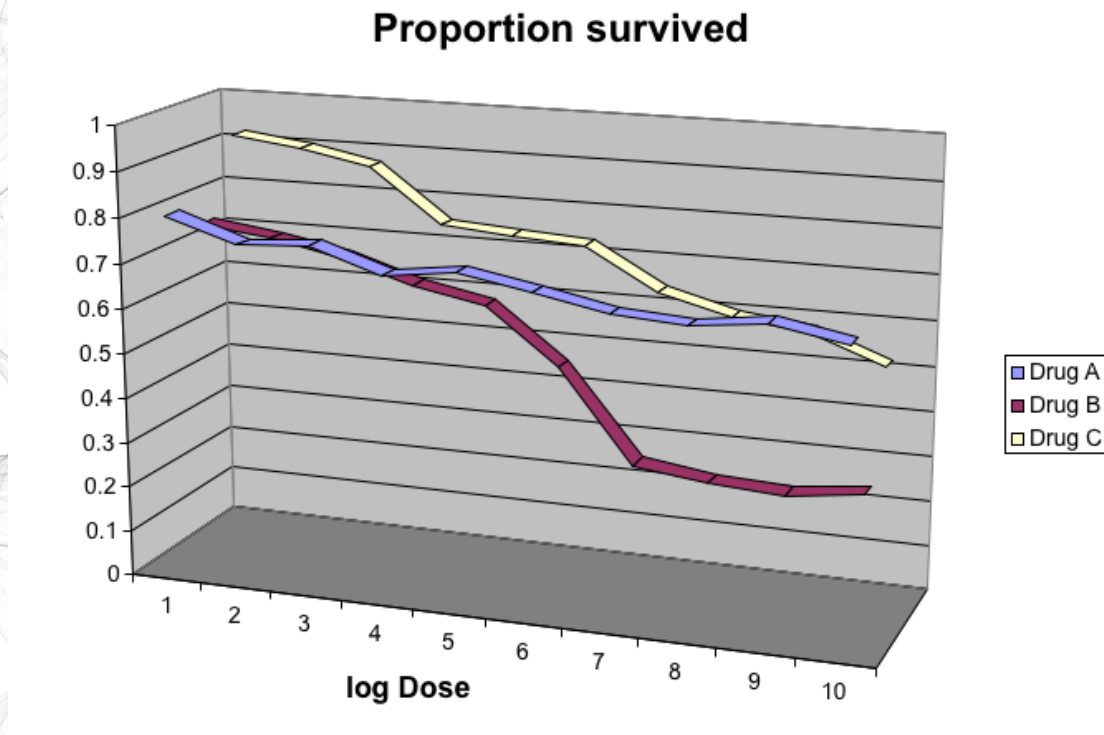


```
R 4.2.2 · ~/
> gapminder |>
+ filter(year == 2010 & !is.na(dollars_per_day) &
+       country %in% country_list) |>
+ mutate(year = factor(year)) |>
+ ggplot(aes(dollars_per_day, life_expectancy, size=population/10^6)) +
+ geom_point(aes(color=group), alpha=0.5) +
+ scale_x_continuous(trans = "log2")
```

The population size is added as the size of dots

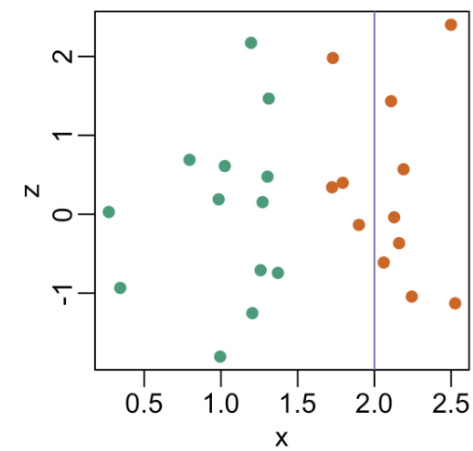
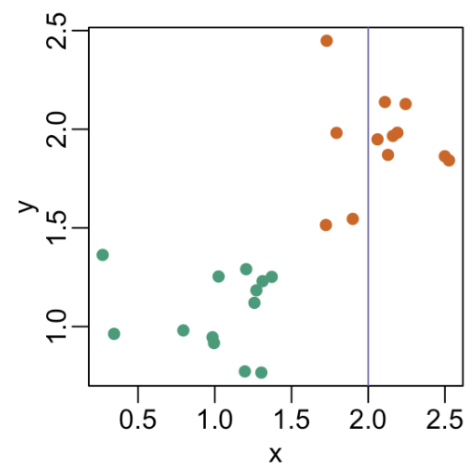
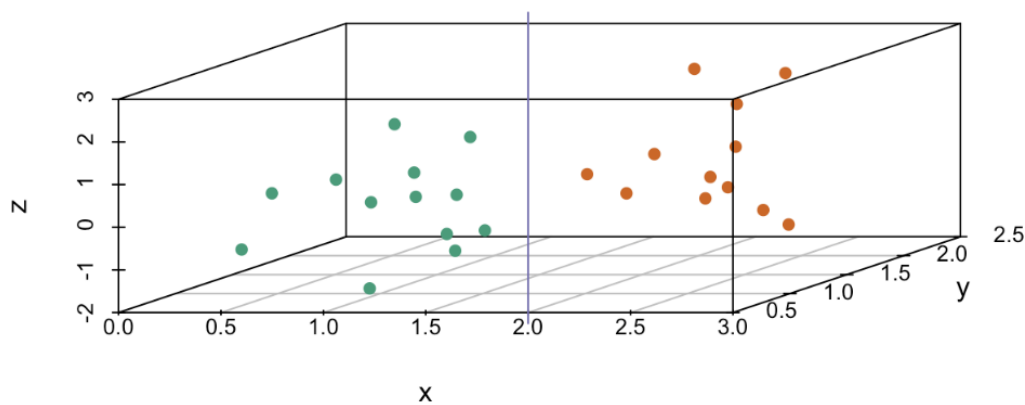
You can add different dot shapes as the fourth variable, but avoid adding more than three variables to a plot unless the data points are scarce

Avoid (pseudo) 3D plots



Just don't use pseudo 3D!

Avoid (pseudo) 3D plots



Use 1 to 3 decimal points, consistently

state	year	Measles	Pertussis	Polio
California	1940	37.8826320	18.3397861	0.8266512
California	1950	13.9124205	4.7467350	1.9742639
California	1960	14.1386471	NA	0.2640419
California	1970	0.9767889	NA	NA
California	1980	0.3743467	0.0515466	NA

Message conveyed: Hey, you know what, I don't really care

```
R 4.2.2 · ~/   
> gapminder <- mutate(gapminder, dollars_per_day=round(dollars_per_day, 2))
```

dollars_per_day
NA
3.4054577
NA
NA
14.3931528
NA
NA
25.7347849
20.3157788
NA
32.6755339
NA
0.7256889
9.3025497
NA



dollars_per_day
NA
3.41
NA
NA
14.39
NA
NA
25.73
20.32
NA
32.68
NA
0.73
9.30
NA

Case study: vaccines and infectious diseases

AD

How to persuade less educated people to get vaccinated?



When it comes to persuading less educated people to get vaccinated, it's important to keep in mind that they may have different concerns and barriers than those with higher levels of education. Here are some tips that may be helpful:

1. Use clear and simple language: Use language that is easy to understand and avoid technical jargon. This will help ensure that your message is accessible to everyone, regardless of their educational background.
2. Address common concerns: Many people may have concerns about the safety and effectiveness of the vaccine, as well as any side effects. Address these concerns in a respectful and informative manner, using trusted sources of information.
3. Share personal stories: Personal stories from people who have been vaccinated can be a powerful way to help others understand the benefits of the vaccine. Encourage those who have already been vaccinated to share their experiences with their family and friends.
4. Emphasize the benefits: It's important to emphasize the benefits of the vaccine, such as protection against serious illness and the ability to return to normal activities. This can help people understand why getting vaccinated is important.
5. Be patient and understanding: Some people may need more time to make a decision about getting vaccinated. Be patient and understanding, and avoid using judgmental language or tactics.
6. Use trusted messengers: People may be more likely to listen to messages about vaccination from people they know and trust, such as their healthcare provider, a community leader, or someone from their religious organization.
7. Address logistical barriers: In some cases, logistical barriers such as transportation or time off from work may be preventing people from getting vaccinated. Offer assistance in finding solutions to these barriers.

Remember that persuading someone to get vaccinated is not a one-size-fits-all approach, and it may take multiple conversations and a range of strategies to be successful. The key is to be respectful, informative, and understanding.

Effective communication of data is a strong antidote to misinformation and fear-mongering

Battling Infectious Diseases in the 20th Century: The Impact of Vaccines

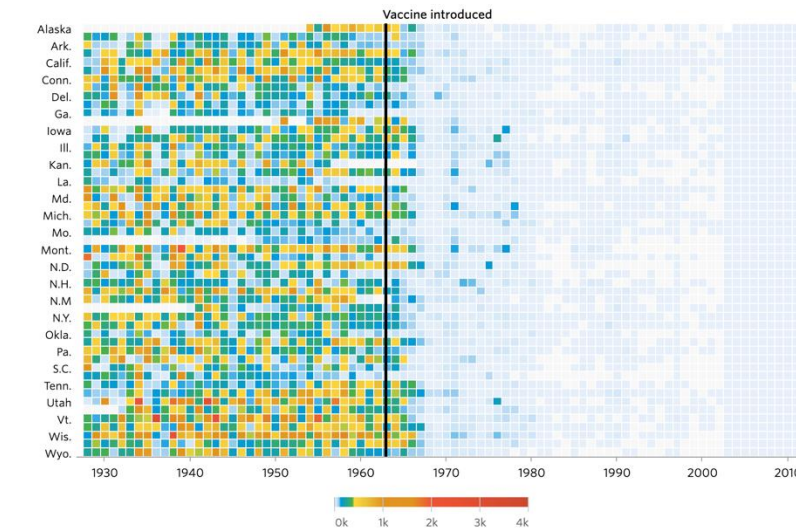
By **Tynan DeBold** and **Dov Friedman**

Published Feb. 11, 2015 at 3:45 p.m. ET

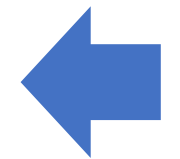
The number of infected people, measured over 70-some years and across all 50 states and the District of Columbia, generally declined after vaccines were introduced.

The heat maps below show number of cases per 100,000 people.

Measles



Note: CDC data from 2003-2012 comes from its Summary of Notifiable Diseases, which publishes yearly rather than weekly and counts confirmed cases as opposed to provisional ones.



Let's try to reconstruct this graph

<http://graphics.wsj.com/infectious-diseases-and-vaccines/>

Get ready

R 4.2.2 · ~/ ↗

```
> library(tidyverse)
> library(RColorBrewer)
> library(dslabs)
> data(us_contagious_diseases)
> names(us_contagious_diseases)
```

[1] "disease"

"state"

"year"

"weeks_reporting"

[5] "count"

"population"

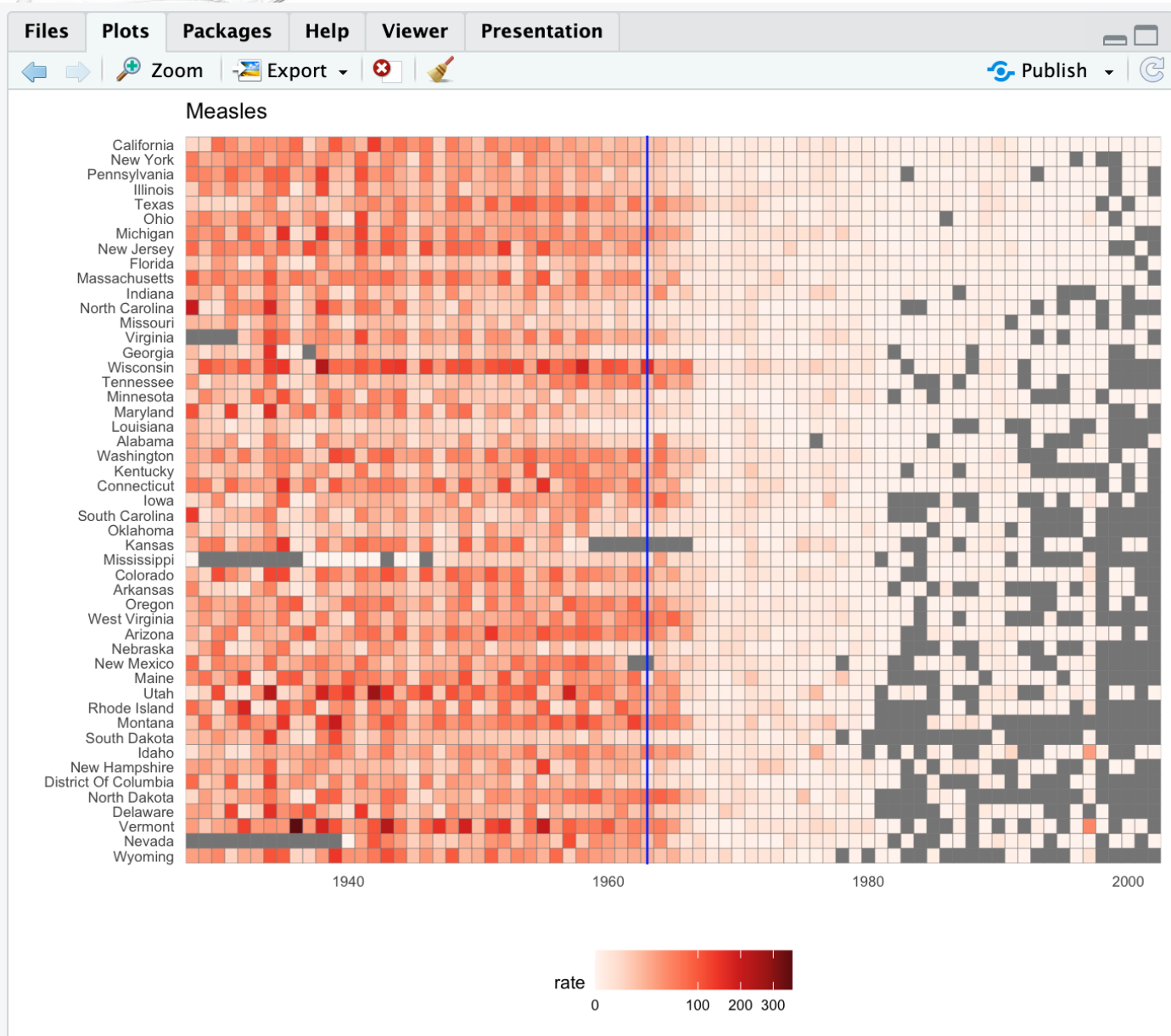


Data wrangling

R 4.2.2 · ~/

```
> the_disease <- "Measles"
dat <- us_contagious_diseases |>
  filter(!(state %in% c("Hawaii", "Alaska"))) & disease == the_disease) |> Exclude Hawaii and Alaska due to insufficient data
  mutate(rate = count / population * 10000 * 52 / weeks_reporting) |> Calculate vaccinate rate with normalization
  mutate(state = reorder(state, population, median, na.rm = TRUE)) Reorder according to median population size along the years
```

Voila!



R 4.2.2 · ~/

```
> dat |> ggplot(aes(year, state, fill = rate)) +  
+ geom_tile(color = "grey50") +  
+ scale_x_continuous() +  
+ scale_fill_gradientn(colors = brewer.pal(9, "Reds"), trans = "sqrt") +  
+ geom_vline(xintercept=1963, col = "blue") +  
+ theme_minimal() +  
+ theme(panel.grid = element_blank(),  
+       legend.position="bottom",  
+       text = element_text(size = 8)) +  
+ labs(title = the_disease, x = "", y = "")
```